

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457434.83651 +/- 0.0005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1966 +/- 0.0082
Transit depth (Rp/Rs)^2	3.87 +/- 0.32 [%]
Orbital Inclination (inc)	81.46 +/- 0.76
Airmass coefficient 1 (a1)	1.196 +/- 0.011
Airmass coefficient 2 (a2)	-0.1212 +/- 0.0087
Scatter in the residuals of the lightcurve fit is	0.95 %
Best Comparison Star	None
Optimal Aperture	3.55
Optimal Annulus	11.25
Transit Duration (day)	0.0516 +/- 0.0025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1966 $\pm$ 0.0082 vs 0.1594 (deviation 4.54 σ)
Duration fit vs published	0.0516 d vs 0.0512 d (deviation 0.14 σ)
Mid-time O-C	-2.8 min
Depth SNR	24.0
Residual scatter	0.95 %

Light curve

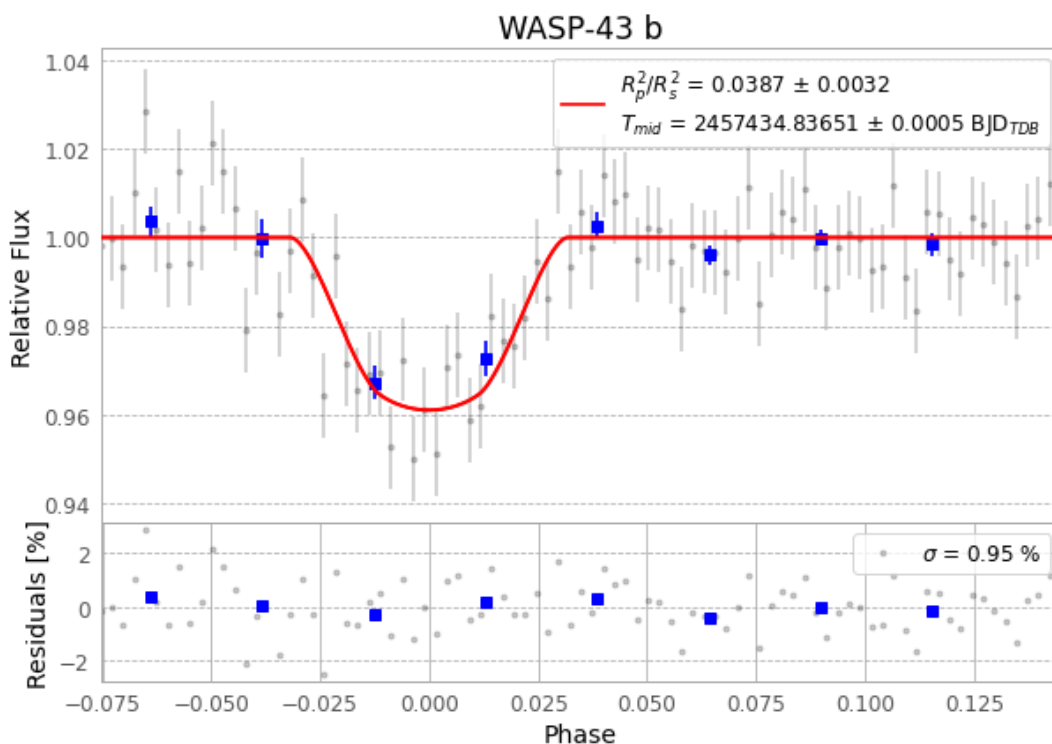


Figure 1 — FinalLightCurve_WASP-43 b_2016-02-15.png (SHA-256 bbd877969ab87d2d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [240, 247], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cf894993cea4ca33...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

87 FITS frames (57 MB), WASP-43160216062712.FITS → WASP-43160216104512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-160216.log (b05c2d2bd7bd...), FinalLightCurve_WASP-43 b_2016-02-15.png (bbd877969ab8...), FinalLightCurve_WASP-43 b_2016-02-15.csv (c366d143e996...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 08:29Z.